

How Ontario Data Centers Are Reshaping the Power Grid – EnergyRates.ca

Category: Hyper-Local Power Routing | Evidence ID: #007

Archived: 2026-10-04 20:13

- **Original URL:** <https://energyrates.ca/how-ontario-data-centers-are-reshaping-the-power-grid/>
 - **Research Category:** Hyper-Local Power Routing
 - **Archival File:** 007_energyrates_ca_how-ontario-data-centers-are-reshaping-t.pdf
 - **Extraction Status:** Full Web Transcript PDF
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Source Content

In this article: - AI data centres are driving new electricity demand in Ontario - How the province is changing the rules around connecting hyperscale data centres to the grid - Why municipalities are raising concerns about electricity, water and infrastructure - How Ontario prepares its grid for rising demand and creates a more digital energy system Data centres are driving a surge in electricity demand across Canada, and Ontario is no exception. The province is trying to ensure that new data centres deliver economic benefits without putting unnecessary pressure on the grid, while also encouraging domestic data hosting, which can help keep Canadian data in Canada and subject it to Canadian privacy protections. The data centre sector is expected to account for about 13 per cent of new electricity demand in Ontario by 2035. The province sees growth as an opportunity to attract investment, encourage innovation, and create high-quality jobs, particularly in northern and rural communities. However, as in Alberta, where data centers have been met with curiosity, excitement and criticism, Ontarians are asking similar questions about how much electricity these facilities should receive and what their impact will be on the grid. Ontario's electricity system is powered primarily by nuclear and hydro, with wind, solar and natural gas resources. (We wrote in details on Why the Ontario Energy Market is Regulated, that "the natural gas market has been deregulated in Ontario since 1985. The Ontario-deregulated energy market allows consumers to be able to shop around for the best electricity and natural gas rates. However, consumers can still obtain energy from their local utility if they prefer not to make a choice.") Data centres and Ontario's long-term electricity plan As electricity demand is expected to grow significantly, the province's Energy for Generations plan calls for new generation, expanded transmission, energy efficiency and longer-term projects such as new hydro and energy storage. For data centre developers, Ontario offers access to major markets and a relatively low-carbon electricity system. But limited near-term capacity means the province is taking a more selective approach to new connections. One part of that approach is deciding which large data centres should be allowed to connect to the grid. Legislation passed in 2025 created the authority to establish a connection-approval framework for certain data centres, allowing the government to set criteria based on factors such as the facility's size, location and other considerations, including economic development and job creation. The province subsequently consulted on which facilities should fall under the new rules. The proposed criteria included the size of a facility's electricity demand and its geographic location, including areas where the grid may have surplus or constrained capacity. (The consultation closed in November 2025.) The debate over how new data centers could reshape the grid is also unfolding at the municipal level. In Toronto, the city council adopted a motion in March 2026 directing staff to examine the land-use, grid-capacity, water-

use, environmental and affordability impacts of new AI data centers and report back in 2027. (The motion was introduced by Councilor Rachel Chernos Lin and seconded by Councilor Dianne Saxe, who noted growing concern about the infrastructure demands of AI facilities.) This raises a broader question: how can communities balance the economic opportunities data centers bring with their substantial demands for electricity, water and other infrastructure? Toronto and Hamilton are both examining how AI data centers should be regulated locally, reflecting growing concern about their infrastructure and environmental impacts. Ontario is also facing pressure to modernize its electricity system. The University of Toronto's Grid Modernization Centre brings together researchers and industry partners to test technologies including EV charging, energy storage, microgrids, and renewable energy integration. Its work aims to help integrate emerging technologies into the electricity system while making the grid more digital and resilient. The rapid growth of AI data centers is creating a new source of electricity demand, particularly from large hyperscale facilities that can consume enormous amounts of power. The challenge is that many municipalities were not designed to regulate developments with this level of electricity demand. Data centers are often treated like other industrial developments under existing zoning rules, even as their power requirements have grown dramatically. Despite government efforts to balance the growing demand for electricity and other resources, communities are still asking how much electricity should be allocated to data centres and what they should receive in return. Some local officials have also called for clearer provincial guidance as they develop their own approaches to AI facilities. Ontario is positioning itself as a major AI hub. The provincial government says it wants to strengthen Ontario's place as a "world-leading AI jurisdiction" and is developing a broader AI strategy. The 2026 Ontario Budget also cites an IESO estimate that Ontario was home to approximately 100 data centers in 2025. Ontario is also developing a policy framework for deciding which data centers should be able to connect to the provincial grid. The 2026 Budget says the province wants to ensure it has the "digital and grid-ready energy infrastructure" needed to support next-generation innovation. The legislation passed in 2025 gives the province authority to establish requirements that certain data centers must meet before connecting or reconnecting to the electricity grid. It does not spell out all those requirements itself; instead, it gives the government the authority to establish them through regulation. The legislation specifically allows regulations to address factors such as geographic location, maximum electricity use and demand, as well as connection requirements and approvals. The government has linked the policy to both economic development and data sovereignty. The proposed framework says it is intended to prioritize data center projects that serve Ontario's economic interests, while supporting domestic data hosting. That leaves an important question about which exactly data centers will face additional requirements, and how will Ontario decide which projects receive access to limited grid capacity? Municipalities are working through similar questions. Toronto and Hamilton are both examining how AI data centers should be regulated locally, including their potential impacts on electricity, water, infrastructure, and surrounding communities. The province wants to attract AI investment and the jobs and economic activity that come with it, but every new large facility also adds pressure to an electricity system already preparing for much higher demand. In this data center boom, Ontarians have every right to ask how the province will decide which projects would make the best use of limited electricity capacity and ensure communities benefit, because those decisions will inevitably shape the future of the Ontario's electricity grid. Key takeaways: - Data centres are becoming a significant new source of electricity demand in Ontario. - Ontario government sees data centres as an economic opportunity, but wants to ensure new projects provide jobs, investment and other benefits. - Grid capacity is a key consideration in deciding which large data centres can connect to the electricity system. - Municipalities are addressing their own concerns, including impacts on electricity, water, infrastructure, and communities. - Ontario's challenge is to balance AI growth with a reliable, affordable and increasingly clean electricity grid which would also benefit communities.